

Project type:	Master's thesis
Project title:	Optimal data curation for self-consuming generative models: a Gaussian analysis
Student:	To be assigned
Supervisors:	Lucia Pezzetti (lucia.pezzetti@ai.ethz.ch) Zhiyu He (zhiyhe@ethz.ch)
Supervising professor:	Prof. Dr. Florian Dörfler
Semester:	To be determined
Start date – End date:	To be determined

Modern generative models are increasingly trained on web-scale datasets that contain content produced by earlier models. When this process is repeated over several generations, errors in the synthetic component can accumulate and the learned distribution can drift away from the original data distribution. This phenomenon, known as *model collapse*, can appear as bias in the mean, shrinking variance, or loss of low-probability modes [8, 1, 2, 5, 4].

This thesis studies a concrete intervention: at each training generation, a curator may inject a limited amount of verified real data. The goal is to decide *when* and *how much* real data to inject so that model drift remains small over a finite horizon while respecting a data-acquisition budget. The project combines stochastic optimal control, optimal transport, probability, and numerical experimentation.

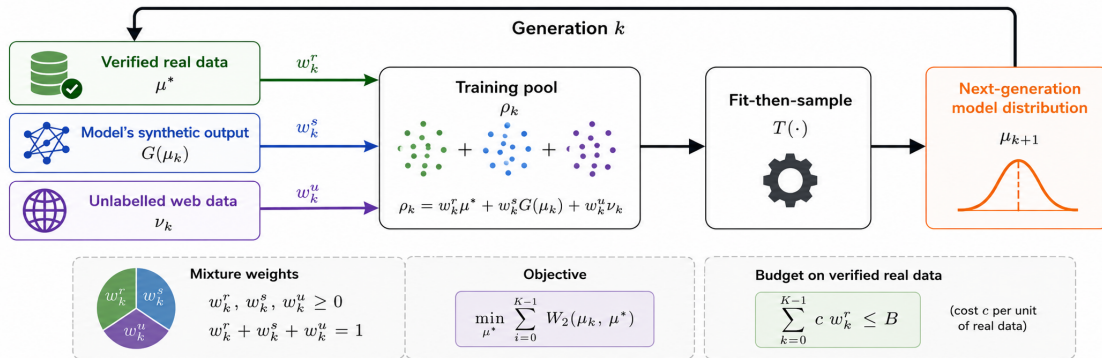


Figure 1: Schematic of the self-consuming loop. At generation k , the curator mixes verified real data μ^* , synthetic data produced by the current model, and unlabelled web data ν_k . A fit-then-sample operator T produces the next model distribution μ_{k+1} . The control variable sets the mixture weights, subject to a budget on verified real data.

Optimal-control formulation. Let $\mu_k \in \mathcal{P}_2(\mathcal{X})$ denote the model distribution at generation k . The curator chooses $u_k = (w_k^r, w_k^s, w_k^u) \in \Delta_2$, where the three components are the fractions of verified real data, synthetic data, and unlabelled web data in the next training set, respectively. A generic controlled loop is

$$\mu_{k+1} = T(w_k^r \mu^* + w_k^s G(\mu_k) + w_k^u \nu_k). \quad (1)$$

For a horizon K and real-data budget B , a representative objective is

$$\min_{u_{0:K-1}} \mathbb{E} \left[\sum_{k=0}^{K-1} W_2^2(\mu_k, \mu^*) \right] \quad \text{s.t.} \quad \sum_{k=0}^{K-1} c w_k^r \leq B. \quad (2)$$

In general, the corresponding Bellman equation is posed on the infinite-dimensional space of probability measures and is not directly tractable.

Scope of the thesis: Gaussian models. The core thesis studies Gaussian target, model, and contamination distributions. This setting is simplified but useful: it captures mean drift and covariance collapse while allowing exact calculations. It also covers classical linear latent-variable models such as probabilistic PCA, factor analysis, and linear VAEs.

Write the model distribution as $\mu_k = \mathcal{N}(m_k, \Sigma_k)$. Under suitable assumptions on the fit-then-sample operator, the loop closes over the first two moments, yielding a finite-dimensional stochastic recursion for (m_k, Σ_k) . Moreover, W_2^2 between Gaussian distributions has the closed form

$$W_2^2(\mathcal{N}(m, \Sigma), \mathcal{N}(m^*, \Sigma^*)) = \|m - m^*\|_2^2 + \text{tr} \left(\Sigma + \Sigma^* - 2((\Sigma^*)^{1/2} \Sigma (\Sigma^*)^{1/2})^{1/2} \right), \quad (3)$$

where the covariance term is the squared Bures distance [10]. The natural finite-dimensional control state is therefore (m_k, Σ_k, b_k) , with b_k denoting the remaining budget. A further reduction to a scalar drift state is possible only under additional structure, for example isotropic or simultaneously diagonalizable covariances; identifying the precise conditions for such a reduction is part of the project.

Policy structure. The project will investigate whether the optimal injection policy is monotone or threshold-based in suitable reduced models, drawing on classical inventory-control theory [7, 9]. These conditions must be verified rather than assumed; a clean result under a clearly stated restricted model, a numerical policy characterisation, or a counterexample are all valid outcomes.

Optional robustness extension. Unlabelled web data cannot generally be certified. If the core objectives are completed on schedule, the project will model ν_k as belonging to a Wasserstein ambiguity set of radius ε and formulate a minimax variant of (2). The aim is to quantify how uncertainty about web-data provenance changes the curation policy and its cost, using tools from Wasserstein distributionally robust optimization where applicable [6].

Project objectives.

Core objectives:

1. **Literature and model specification.** Review model-collapse mechanisms and Gaussian Wasserstein geometry. Specify the finite-sample training protocol, including mixture rules, estimation noise, and the fit-then-sample operator.
2. **Moment dynamics and sufficient statistics.** Derive the recursion for (m_k, Σ_k) and identify the assumptions under which it is exact. Recover known constant-control collapse behaviour as a consistency check [2, 5].
3. **Dynamic programming and structural analysis.** Formulate the finite-horizon Bellman recursion on (m_k, Σ_k, b_k) . Analyse convexity, monotonicity, and possible state reductions. Investigate policy structure for tractable settings, for example isotropic models, via analytical arguments and numerical policy maps.
4. **Baselines and numerical validation.** Implement the controlled Gaussian loop and compare the optimal policy against no-injection, fixed-fraction, and accumulate-all baselines across budget B , horizon K , and sample size.
5. **Reporting.** Produce documented code, a final thesis report, and a seminar presentation.

The thesis is considered successful upon delivery of objectives 1–5. The extensions below are not required, but would constitute additional contributions.

Optional extensions:

6. **Distributionally robust curation.** Model ν_k inside a Wasserstein ambiguity set, derive the minimax control problem [6], and estimate the price of robustness.

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7. **Beyond Gaussians.** Test whether insights transfer to a small VAE or diffusion model using sliced Wasserstein as a drift proxy.

Timeline.

- **Month 1:** Literature review; model specification; uncontrolled baseline simulator.
- **Month 2:** Derive and validate the mean–covariance recursion; recover constant-control collapse.
- **Month 3:** Formulate the Bellman recursion; implement solutions for isotropic models.
- **Month 4:** Policy structure analysis; comparison against simple baselines.
- **Month 5:** Complete numerical study; begin optional extensions if core is done.
- **Month 6:** Final report, documented code, and seminar presentation.

Qualifications. We welcome students with solid foundations in linear algebra and probability, and experience programming in Python. Background in stochastic processes, optimal control, or machine learning is helpful but not required.

How to apply. Please send your CV (including a transcript of records and a brief statement of motivation) to lucia.pezzetti@ai.ethz.ch and zhiyhe@ethz.ch.

Supervision. This project is supervised within the Automatic Control Laboratory (IfA) at ETH Zurich and the ETH AI Center.

Procedures.

1. The student and supervisor will follow the checklists “*Procedures for Students*” and “*Procedure for Supervisors*”, available from the *Internal* section of the Automatic Control Laboratory website.
2. At the end of the project, the student must submit a final report no longer than 50 pages and give a 20-minute seminar at the Automatic Control Laboratory.

References

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- [3] Apratim Dey and David Donoho. Universal laws for model collapse via data mixing and data contamination. 2024.
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- [5] Matthias Gerstgrasser, Rylan Schaeffer, Apratim Dey, Rafael Rafailov, Sanmi Koyejo, David Donoho, and Suriya Goel. Is model collapse inevitable? breaking the curse of recursion by accumulating real and synthetic data. 2024.
- [6] Peyman Mohajerin Esfahani and Daniel Kuhn. Data-driven distributionally robust optimization using the Wasserstein metric: Performance guarantees and tractable reformulations. *Mathematical Programming*, 171(1–2):115–166, 2018.
- [7] Herbert Scarf. The optimality of (s, S) policies in the dynamic inventory problem. *Mathematical Methods in the Social Sciences*, pages 196–202, 1960.
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- [10] Cédric Villani. *Optimal Transport: Old and New*, volume 338 of *Grundlehren der mathematischen Wissenschaften*. Springer, 2009.